- **6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables**
  - **Analysis:** This is the first and most critical step in problem-solving. Students need to identify core information from the text description: initial penguin population (10,000), birth rate (1 chick per couple), death rate (20%), and the time frame (end of one year). This requires converting the real-world scenario into actionable mathematical elements.
- **7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
  - **Analysis:** Students need to break down the complex one-year change process into smaller, manageable steps: first calculate the number of births, then add the births to the initial number, and finally calculate and subtract the number of deaths. This is a simplification process that allows the problem to be solved through simple arithmetic operations.
- **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
  - **Analysis:** Students must recognize that this is a word problem involving **percentage increase** (birth rate, although given in terms of couples) and **percentage decrease** (death rate), allowing them to call upon the appropriate mathematical procedures.
- **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
  - **Analysis:** This ability emphasizes understanding textual assumptions, such as "5,000 couples" implying that 5,000 chicks are born. Accurately interpreting this information is a prerequisite for building the correct model.

- **6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables**
  - **分析：** 这是解决问题的第一步，也是最关键的一步。学生需要从文本描述中识别出核心信息：初始企鹅数量（10,000只）、出生率（每对夫妇1只）、死亡率（20%）以及时间范围（一年结束时）。这要求将现实情境转化为可操作的数学元素 (p. 1)。
- **7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
  - **分析：** 学生需要将复杂的一年变化过程分解为更小的、可管理的步骤：首先计算出生数量，然后将出生数量加到初始数量上，最后计算死亡数量并减去。这是一个简化的过程，使得问题可以通过简单的算术运算解决。
- **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
  - **分析：** 学生必须认识到这是一个涉及**百分比增加**（出生率，尽管是以夫妇形式给出）和**百分比减少**（死亡率）的应用题，从而能够调用适当的数学程序。
- **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
  - **分析：** 这项能力强调理解文本假设，例如“5,000对夫妇”意味着有5,000只小企鹅出生 (p. 1)。准确解读这些信息是建立正确模型的前提。